Reconcilable Differences: Booting Linux on a PowerVM LPAR

George Wilson, Nayna Jain, Avnish Chouhan, Nageswara Sastry
IBM Linux Technology Center

Linux Plumbers Conference

December 13, 2025





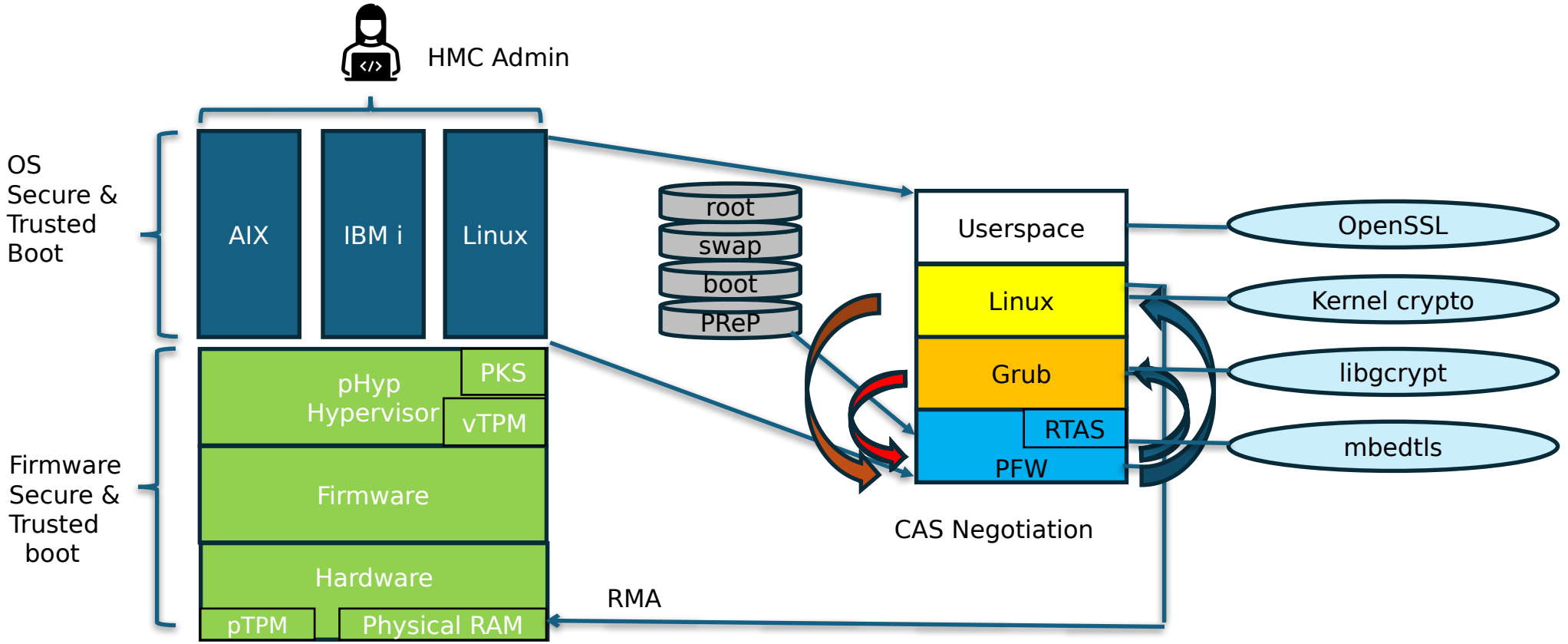
LPAR Boot Stack

- IBM Power servers run PowerVM, a Type I hypervisor, which provides an abstraction layer on top of the HW
- Logical partitions – LPARs – are strongly separated VMs in which OSs execute
- Partitioning and I/O topology are managed by the Host Management Console (HMC)
- Each LPAR has an Open Firmware (OF) instance called Partition Firmware (PFW) that is an IBM Power specific Open Firmware (IEEE-1275) implementation
- OS boot is initiated by PFW by loading a client program

LPAR Boot Stack

- OF variables control the boot device selection
- The OF client program binary – currently GRUB core.elf – resides in a special well-known partition, the PReP partition, which has no filesystem; or, it is downloaded over TFTP in the case of net boot
- Trusted boot measures and creates boot artifacts, and secure boot verifies the client program before handing over control to it
- Client Architecture Support (CAS) characteristics are exchanged between GRUB and pHyp, and subsequently between the kernel and the pHyp, including Real Mode Area (RMA) size, page table type, endianness, and more

LPAR Boot Stack



Power LPAR Secure Boot vs UEFI

- The system is always virtualized; OS boot is always in an LPAR
- Live Partition Mobility (LPM)
  - In dynamic key management mode, the PKS data migrates along with the other LPAR data
  - Placement must be considered: Is PKS available? Is the vTPM available?
- Different and generally simpler secure boot model
  - Authenticated variables are stored in Platform Keystore (PKS) vs NV-RAM
  - No SMM or SMIs
  - No OOB management of authenticated variables
  - No central secure boot signing authority
  - No key authority transition so there is no shim
  - Either IBM provides the certificates or the sysadmin takes ownership of the PK and signs updates

Power LPAR Secure Boot vs UEFI

- Static vs dynamic certificate management modes
  - By default, the LPAR uses static certificates
  - Static mode was originally developed to meet Common Criteria OSPP and now provides out-of-the box secure boot for Tier 1 Linux distributions
  - HMC physical presence is required to change modes from static to dynamic
  - When switched, existing static certificates become default certificates
- Signing: The ELF vs PE/COFF binary loaded by PFW uses appended signatures borrowed from Linux kernel module signing
- Secure Boot Variables
  - Authenticated variables are directly modeled on Tianocore variables
  - However, they do NOT have the same semantics

Current Considerations

- Power systems have added efficient KVM support for L2 guests fairly recently
  - However, there aren't the same secure boot and other integrity guarantees for L2 guests as there are for L1 guests
  - We could and should add secure boot to QEMU but that only gets us so far because there's a great deal of code to trust in the OS
  - IMA-appraisal - signing and verifying L1 OS files - helps, and is doable out of the box nowadays with RHEL
  - Is IMA sufficient? Does L1 instead need to be a completely immutable base? Then patching is problematic if you want to minimize maintenance downtime. IMA could also verify L2 OS images, potentially obviating the need for L2 secure boot.

Current Considerations

- PQC affects all aspects of our secure boot and trusted boot implementations
  - IETF standards for X.509 and CMS are in the works
  - Making their way into OpenSSL
  - Libgcrypt is currently required for GRUB; would like to transition to PFW interface
  - Mbed TLS is a dependency for PFW
  - Of course, trusted boot must wait for TCG standards and a new TPM
  - Thoughts on a realistic timeline for PQC secure boot across architectures?

Current Considerations

- Mbed TLS
  - Used for PFW verification of GRUB
  - Small community that necessarily moves slowly
  - ML-DSA on roadmap for 2026
  - Beginning more interactive engagement with their developers
  - Does anyone here contribute to Mbed TLS? If so, do you have insights on working with them?
- Multiple signatures
  - Currently multiple signers within a single CMS
  - Considering a CMS array a la PE/COFF if we need more sophisticated composition of signature verifications
  - Does anyone have thoughts on composition of multiple signature verification beyond an always-OR policy?

Current Considerations

- PPC Trusted Computing measurements vs PC Client and UEFI specifications
  - Power servers are neither PCs nor do they use UEFI
  - Some event types are common with UEFI and many are not
  - Reusing UEFI events where it makes sense to do so
  - We reserved an IBM PowerPC event type space with TCG, documented in the *Registry of Reserved TPM 2.0 Handles and Localities*, and we're allocating non-common event types from that space
  - However, this means that attestation servers must understand the vendor-specific event type space
  - Does this seem like a reasonable approach?

Current Considerations

- RMA memory management
  - Hopefully solved for the time being
  - Article: https://www.ibm.com/docs/en/linux-on-systems?topic=servers-memory-consumption-in-grub
  - Can currently boot a large (~500M) 64-bit LSB ELF kernel
  - OF variable settings are required
  - Thoughts on maximum image size?

Current Considerations

- Items for nmbl/UKI
  - CAS negotiation reboots
  - ESP equivalent
  - OF/ELF Systemd boot stub
  - Image discovery and selection
  - Thoughts on reality of nmbl or UKI replacing GRUB and timeline?
- TFTP
  - Has size limitations
  - Considering HTTP and HTTPS
  - How messy is server certificate distribution? Mutual authentication is even messier but could be used for boot entitlement and limiting server load.